Context-Aware LLM Routing Framework for Cloud-Deployed Applications using AI Agents

Optimizing AI applications by dynamic allocation of LLMS

Cloud computing term project
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The Problem: "Compute Overkill"

Monolithic Inefficiency: Current cloud AI relies on a single, massive LLM (e.g., Llama-3-70B) for every query

The Bottleneck: Using a 70B parameter model for simple data extraction wastes significant GPU VRAM and increases latency

Homogeneous Bias: Systems using one model inherit its specific biases, preventing robust error correction

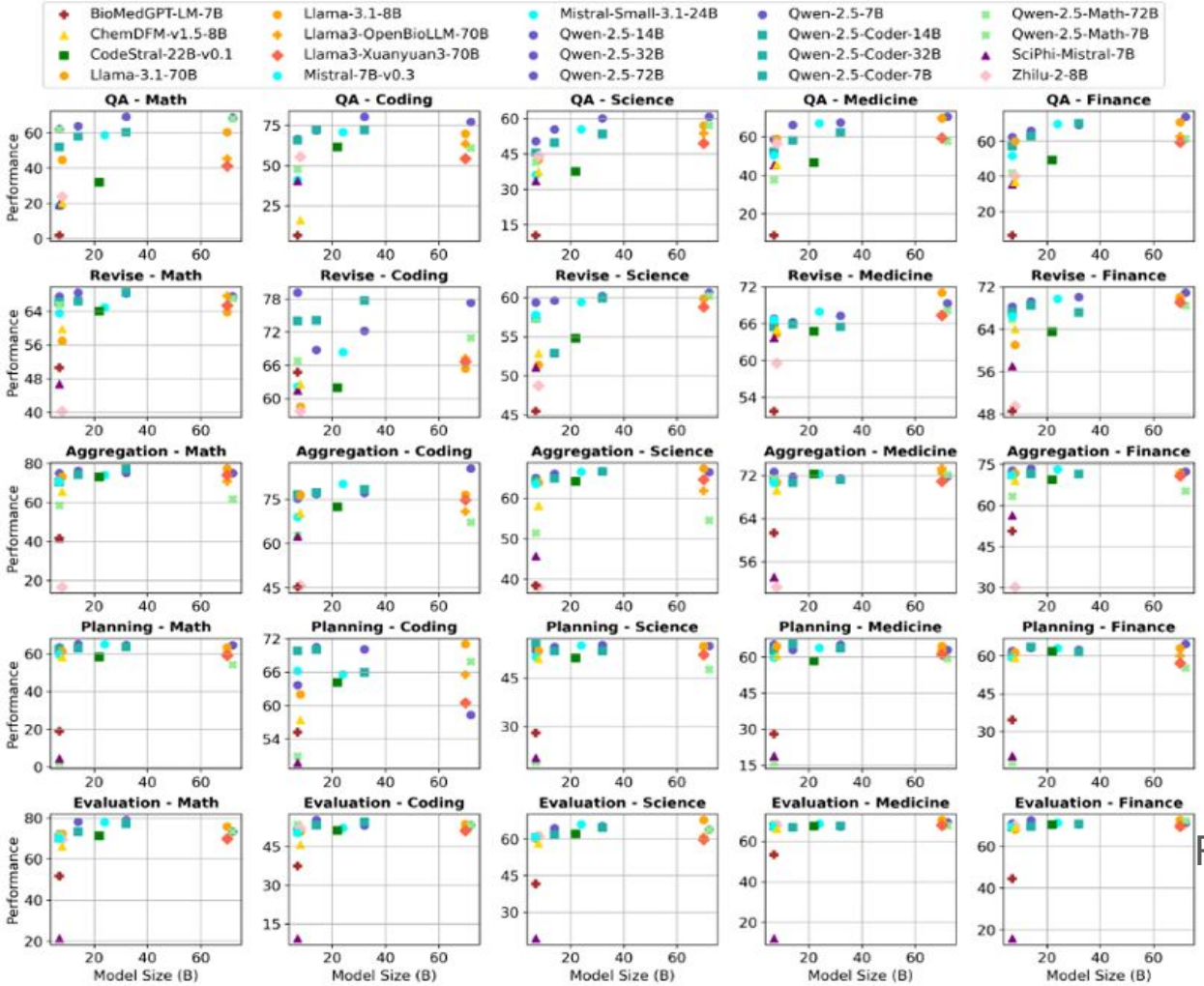
Objective: Intercept queries and route them to the most "economical" yet "capable" model based on intent

Foundations: X-MAS & Model Heterogeneity

Core Principle: No single LLM excels universally across all domains (Math, Law, Code) and functions (QA, Planning)

Heterogeneous Advantage: Leveraging diverse LLMs enhances "collective intelligence," mirroring a team of human specialists

Specialized Efficiency: Smaller, specialized models (e.g., Qwen-Math) often outperform general-purpose frontier models in their specific niche.



Results from the X-Mas paper

Methodology: The Agentic Router

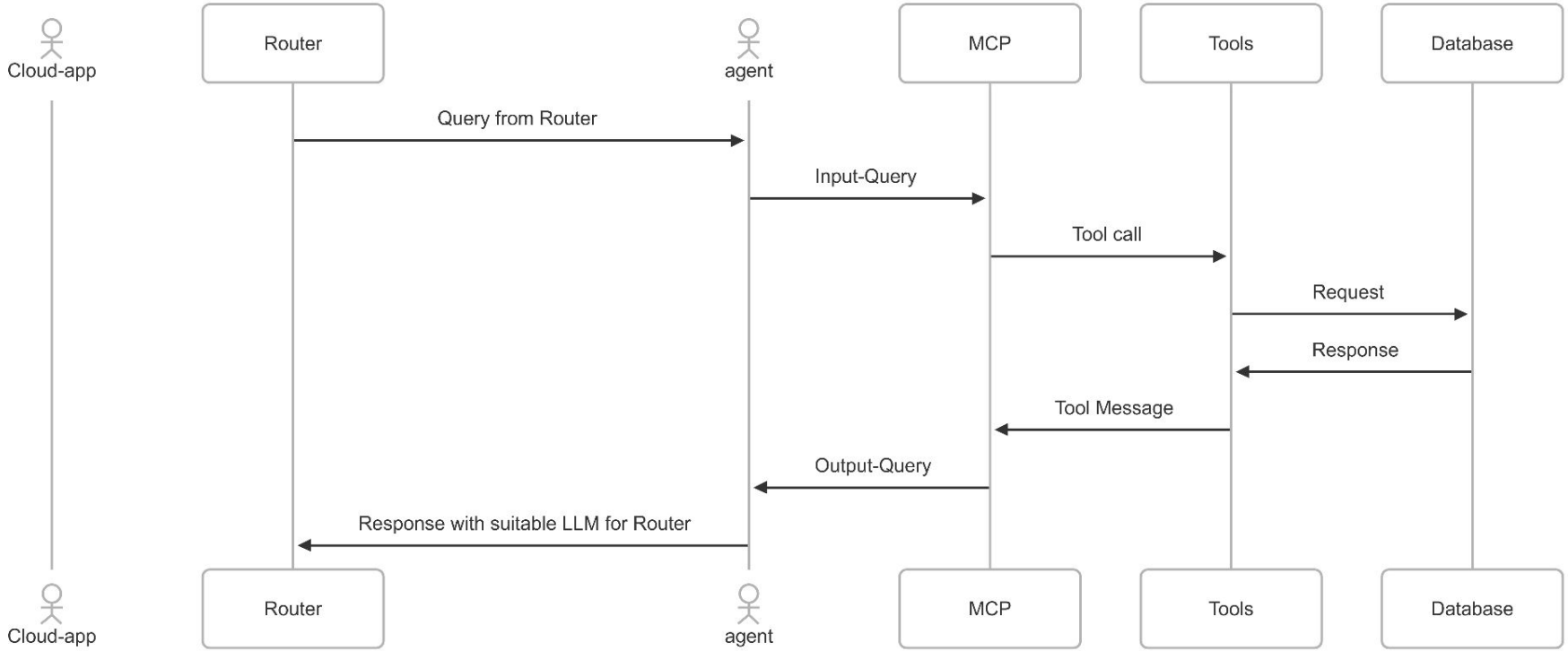
Orchestration: A stateful graph architecture built with **LangGraph**.

Dynamic Selection Process:

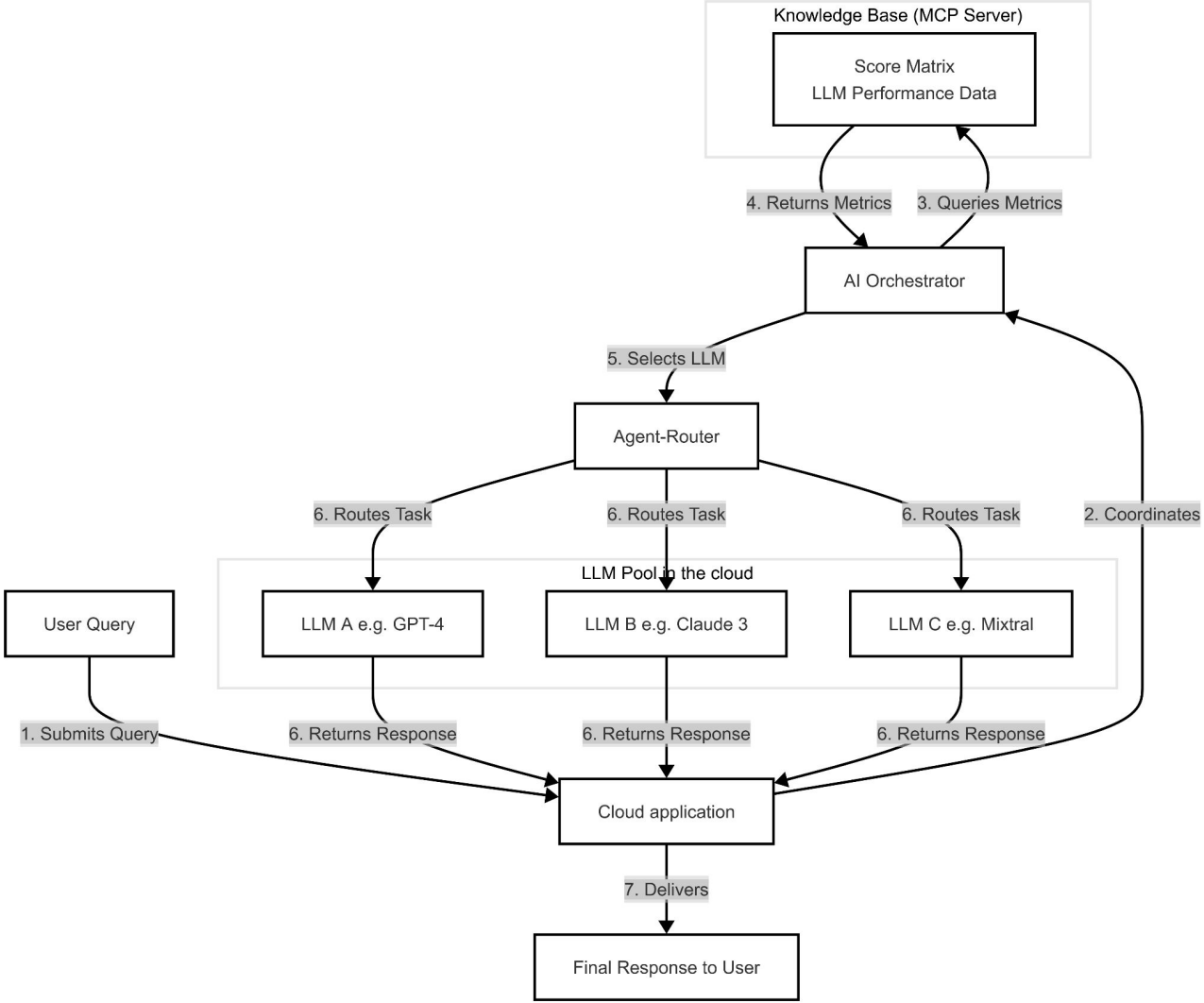
1. **Intent Classification:** Mapping unstructured queries to specific domains (e.g., "Law").
2. **MCP Metric Retrieval:** Fetching real-time benchmarks (Accuracy, Latency, Memory) via the **Model Context Protocol**.
3. **Weighted Scoring:** Calculating the optimal model using the weighted scores.

$$\mathbf{Score} = w_{\mathbf{acc}}(\mathbf{Acc}) + [w_{\mathbf{lat}}(\mathbf{Lat}) + w_{\mathbf{mem}}(\mathbf{Mem})]$$

Where does this agent sit and operate for cloud application?



Agent workflow



Working demonstration

```
Fetching tools from MCP server...
Unclosed client session
client_session: <aiohttp.client.ClientSession object at 0x7f18b0615dc0>
Unclosed client session
client_session: <aiohttp.client.ClientSession object at 0x7f18b0616480>

Invoking Graph with query: 'Give the metrics for History for al models'
DEBUG: Entering node INIT
DEBUG: helper function 'classify_intent'
DEBUG: classify_intent received query: 'give the metrics for history for al models'
DEBUG: fetching models/domains/subdomains
DEBUG: Entering node collect_metrics_step
DEBUG: helper function 'map_query_to_domain_subdomain_llm'
DEBUG: Raw LLM Routing Output:
Domain: General-Knowledge
Subdomain: history
DEBUG: mapping result -> domain=General-Knowledge, subdomain=history
DEBUG: Calling metrics for Llama3.2_1B in General-Knowledge/history
DEBUG: Calling metrics for Qwen2.5_1.5B in General-Knowledge/history
DEBUG: helper function 'select_best_llm'
DEBUG: weighted score: 0.300 for Llama3.2_1B_General-Knowledge_history
DEBUG: weighted score: 0.700 for Qwen2.5_1.5B_General-Knowledge_history
DEBUG: Best model: Qwen2.5_1.5B_General-Knowledge_history
Reason: Selected Qwen2.5_1.5B_General-Knowledge_history based on weighted metrics
DEBUG: Entering node respond_step


--- FINAL OUTPUT ---
Based on the analysis and context:
Context: Selected Qwen2.5_1.5B_General-Knowledge_history based on weighted metrics
...
  eval_duration_ms: 10225.18


Recommended Model for routing: Qwen2.5_1.5B
```

Results & Future Directions

Simulated Gains:

- **Memory:** Significant reduction in peak RAM usage per query by bypassing "frontier" models for routine tasks.
- **Latency:** Selected models which consume less time for generating same results

Conclusion: Dynamic routing is essential for sustainable, production-grade AI.

Future Work: Deploying onto edge devices (Raspberry Pi/Smartphones).

- Enhanced explainability for autonomous model selection decisions